INTRODUCTION

Every cycle tells a story.

- GR05 · Predictive Maintenance
- Aircraft engine Remaining Useful Life
- Selçuk Ata Aksoy · Denzel Gregory Pragassa
- HFU · Industrial Analytics · WS 2026/27

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

INTRODUCTION

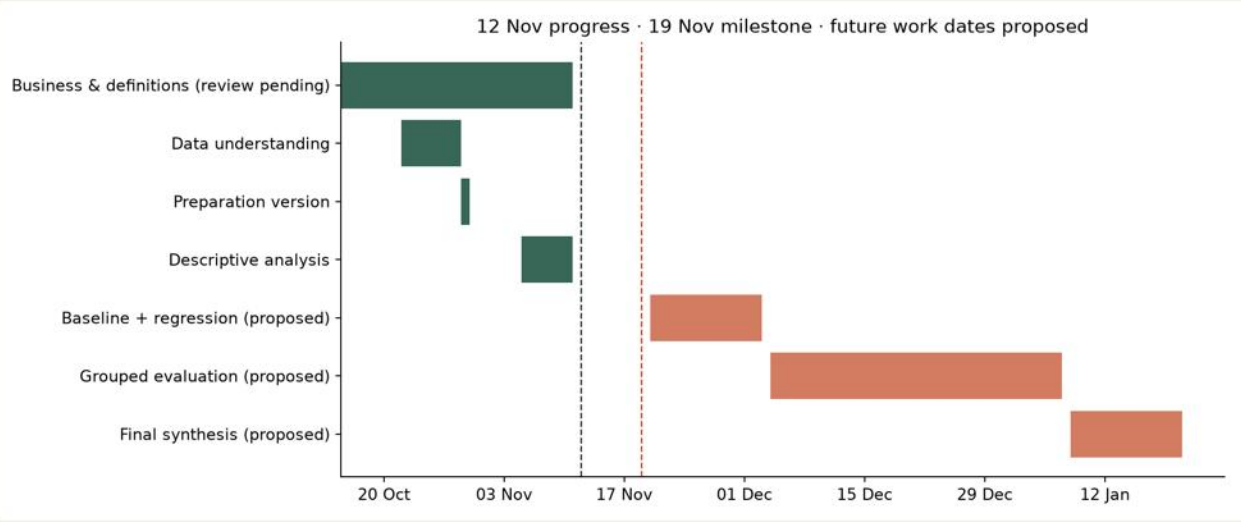
Our question and scope

- Can simple regression estimate RUL better than a transparent baseline?
- Prediction point: end of each completed flight.
- Scope: local DS01 + DS02; offline simulated-data study.
- Deliverable now: explainable evidence and prepared inputs.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

PROJECT PLAN

From question to evidence



- October: Tasks 1–4 → definition, business, understanding, preparation.
- 5–11 Nov: Task 5 → descriptive analysis, review and completion.
- 12 Nov: progress/preparation; 19 Nov: milestone presentation.
- Next: baseline → regressors → grouped evaluation → 21 Jan final.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

DELIVERABLES

What exists and what comes next

- Executed notebooks: understanding, preparation, descriptive analysis.
- Versioned local inputs, fitted preprocessing, audit CSVs and charts.
- Reports and local static pages: findings, preparation, analysis dashboard.
- Pending: human review, model training, MAE/RMSE and business evaluation.

THEORY 01 / 10

Why maintenance planning is uncertain

- Equipment condition changes during operation.
- Planners balance intervention lead time against continued operation.
- Sensor records can inform that decision, but uncertainty remains.

Evidence: Arias Chao et al. (2021), Data 6(1), 5. https://doi.org/10.3390/data6010005

THEORY 02 / 10

Three maintenance strategies

- Reactive: act after a fault or failure is detected.
- Preventive: intervene at scheduled intervals.
- Predictive: use condition/history to estimate future needs.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

THEORY 03 / 10

Degradation is a process

- A history records evolving condition rather than isolated failures.
- Different engines can have different initial health and lifetimes.
- Operating profiles can affect the signals we observe.

Evidence: Arias Chao et al. (2021), Data 6(1), 5. https://doi.org/10.3390/data6010005

THEORY 04 / 10

Condition monitoring: what is observed?

- Flight context: altitude, speed, throttle and inlet temperature.
- Response: temperatures, pressures, shaft speeds and fuel flow.
- Together they describe the engine operating under a mission.

Evidence: Arias Chao et al. (2021), Data 6(1), 5. https://doi.org/10.3390/data6010005

THEORY 05 / 10

Diagnosis and prognosis answer different questions

- Diagnosis: what condition or fault may exist now?
- Prognosis: how might condition evolve, or how long remains?
- GR05 targets prognosis through continuous RUL regression.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

THEORY 06 / 10

RUL is a countdown in flight cycles

- Observed label: final cycle − current cycle.
- Example: final 100, current 30 → RUL 70 cycles.
- Final cycle is used to audit y, never supplied in X.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

THEORY 07 / 10

A prediction needs a clock

- At flight end: the current completed flight is known.
- Previous flights are known; future flights are not.
- Our summaries and rolling features respect that boundary.

Evidence: Task 4 manifest: task4-v1-f8ab427897d5; docs/06-data-preparation.md.

THEORY 08 / 10

The two directions of error matter

- Optimistic RUL: estimates more remaining life than the label.
- Pessimistic RUL: estimates less remaining life than the label.
- Possible implications: delayed vs premature maintenance.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

THEORY 09 / 10

CRISP-DM makes the work traceable

- Business → data understanding → preparation.
- Modeling → evaluation → business implications.
- Each stage can send us back to improve an earlier decision.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

THEORY 10 / 10

Context shift challenges generalization

- A signal can change with mission as well as degradation.
- DS02 development has class 3; test also has classes 1 and 2.
- A result on repeated flights may not generalize to unseen engines.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

BUSINESS 01 / 05

Who would use the output?

- Maintenance planners: lead time and intervention timing.
- Reliability engineers: degradation patterns and error interpretation.
- Operations: downtime and premature-maintenance trade-offs.
- Our users today: students and lecturer reviewing evidence.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

BUSINESS 02 / 05

The proposed value is decision support

- Input: observed operating/sensor histories.
- Output later: an RUL estimate with documented error.
- Potential use: inform planning discussions.
- Value remains hypothetical until validated independently.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

BUSINESS 03 / 05

How will success be assessed?

- MAE: typical absolute error in flight cycles.
- RMSE: highlights larger cycle errors.
- Compare against constant training-RUL baselines.
- Also show per-engine and macro-average errors.

Evidence: https://scikit-learn.org/stable/modules/model_evaluation.html#regression-metrics (accessed 11 Nov 2026).

BUSINESS 04 / 05

Make the scope feasible for two students

- Start with simple baselines and classical regression.
- Use the 20-column baseline selection before 50 candidates.
- Evaluate on separate engine groups.
- No deep learning or live aircraft deployment in initial scope.

BUSINESS 05 / 05

Team responsibilities: proposed, not assumed

- Selçuk: present data preparation and linear regression overview.
- Denzel: present descriptive findings and random forest overview.
- Both: review business claims, rehearse and understand all evidence.
- Confirm presenter allocation and individual contributions together.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

DATA SOURCE

Two local subsets, one lineage

- NASA N-CMAPSS → Kaggle redistribution → local HDF5.
- DS01-005 and DS02-006; approximately 4.958 GiB combined.
- 2020 local preprint differs from 2021 journal description.
- Current source labels recorded; original download-version match unproven.

Evidence: Arias Chao et al. (2021), Data 6(1), 5. https://doi.org/10.3390/data6010005

DATA 01 / 05

Reading → flight → engine history

- 14,159,058 ordered readings.
- 1,542 dataset-engine-flight pairs.
- 19 dataset-scoped source histories.
- Across-subset independence is not established.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

DATA 02 / 05

47 stored fields have different roles

- W 4 + X_s 14: physical channel candidates.
- X_v 14 + T 10: hidden or virtual simulator quantities.
- Y 1: target; A 4: metadata.
- Stored fields are not all legitimate predictors.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

DATA 03 / 05

A number can still be a category

- Unit ID: nominal label, despite float64 storage.
- Flight class: ordered category (short / medium / long).
- RUL: discrete ratio count; absolute temperature: continuous ratio.
- Units and statistical meaning guide valid comparisons.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

DATA 04 / 05

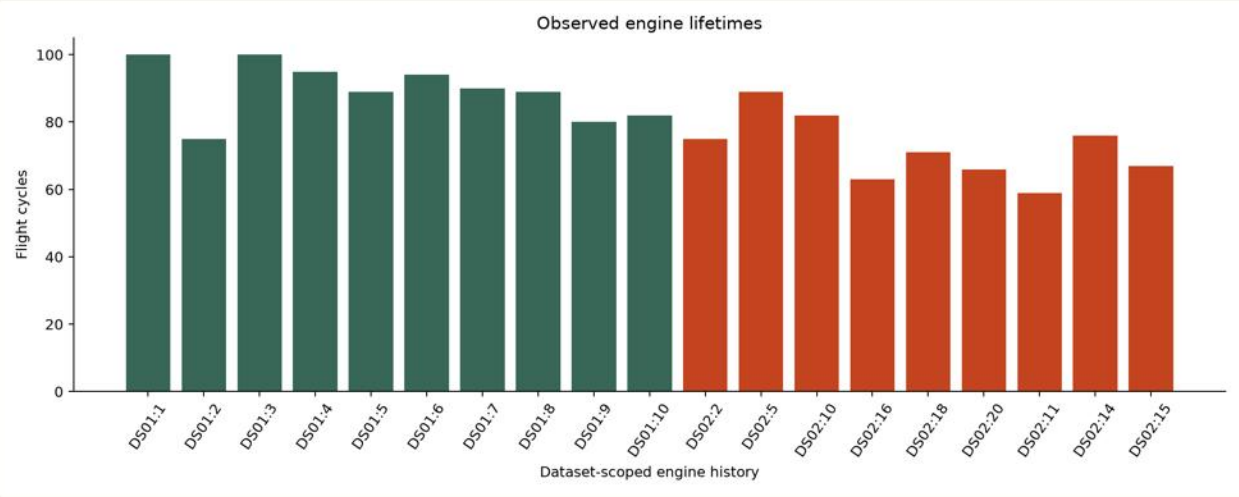
Quality checks passed—with limits

- No NaN/inf in the full Task 3 numeric scan.
- RUL countdown and cycle logic checks passed.
- Adjacent and sampled exact-duplicate checks found zero.
- No proof of physical accuracy or all non-adjacent uniqueness.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

DATA 05 / 05

Engine lifetimes differ

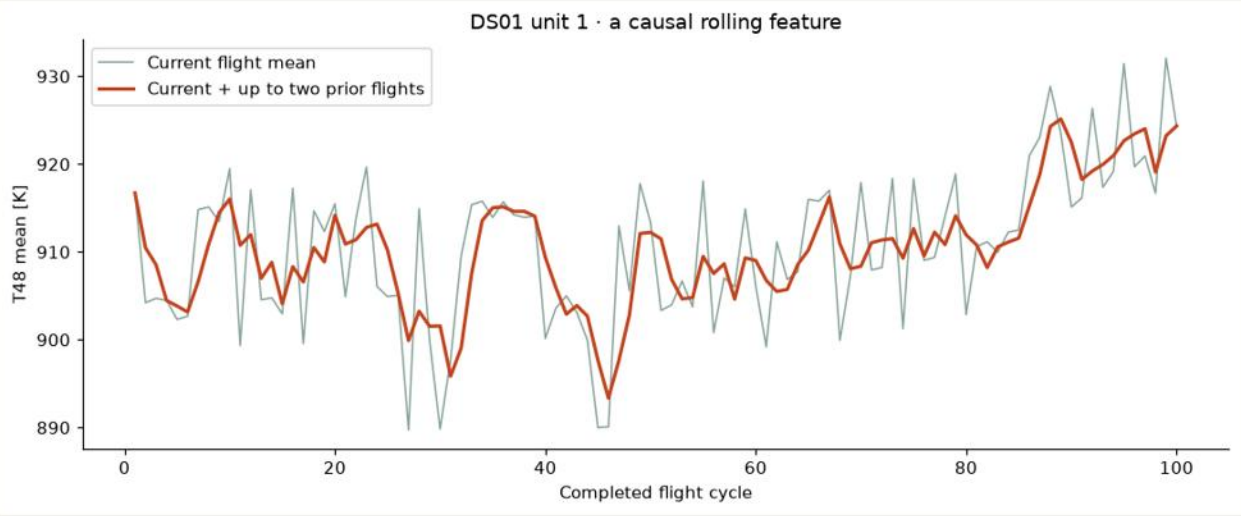


- DS01 histories: 75–100 flight cycles.
- DS02 histories: 59–89 flight cycles.
- Longer missions create more readings, not more engines.

Evidence: Task 3 executed notebook + reports/results/task3_summary.json; figures generated from local DS01/DS02.

PREPARATION 01 / 03

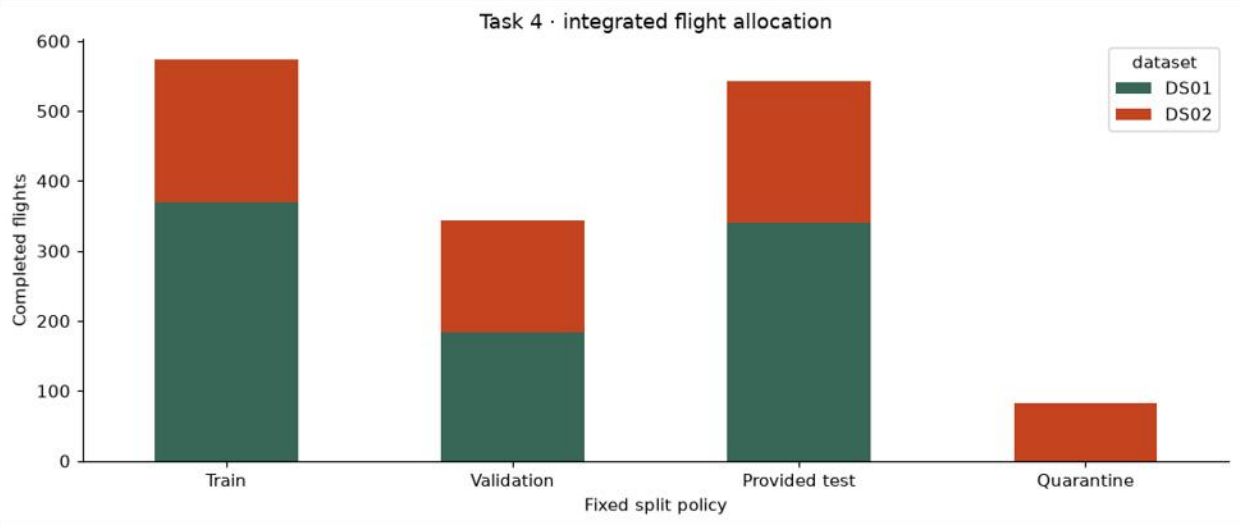
From snapshots to one completed flight



- 17 channel means + 17 within-flight spreads + reading count.
- 3 causal trailing means + 3 previous-flight differences.
- 4 physical combinations; 45 numeric candidates.
- Physical units: m, K, kPa, kg/s, rpm.

PREPARATION 02 / 03

Split first; learn transformations on training only



- 574 train · 343 validation · 543 provided test.
- 82 DS02 unit-10 flights quarantined due to possible DS01 test link.
- Training medians + missing indicators; training standardization.
- No validation/test fitting; shared IDs grouped conservatively.

Evidence: Task 4 manifest: task4-v1-f8ab427897d5; docs/06-data-preparation.md.

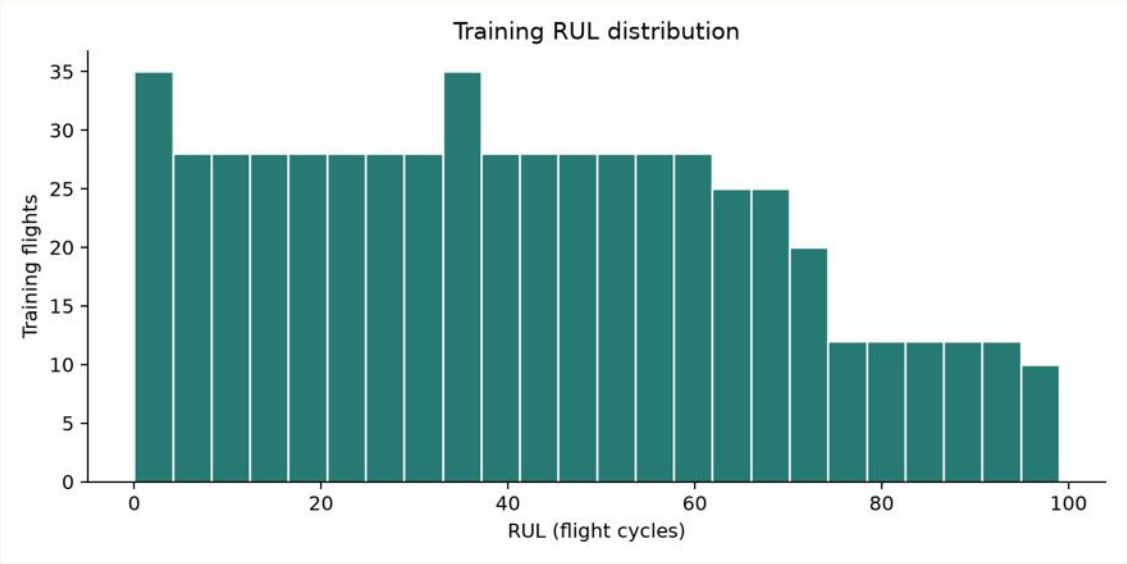
PREPARATION 03 / 03

Raw and prepared are both available locally

- Raw HDF5: immutable originals, excluded from Git.
- Prepared CSVs: X inputs, y RUL, matching row identities.
- 50-column candidate matrices; 20-column baseline option.
- Hashes, config, code and package versions identify the version.

DESCRIPTIVE 01 / 05

The hypothesis-building sample

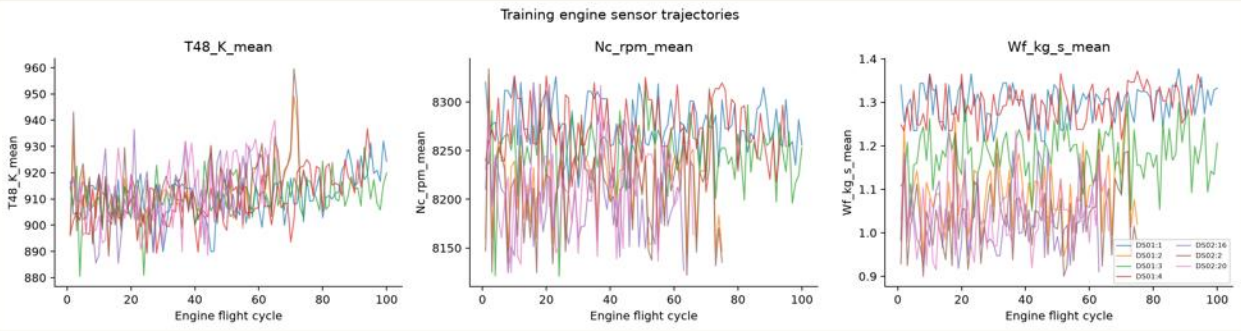


- 574 flights · 7 source histories · 6 conservative groups.
- Training RUL: 0–99 cycles; mean 41.83.
- No test/quarantine sensor hypotheses.
- Validation is a descriptive check already seen in review.

Evidence: Executed Task 5 notebook; reports/results/task5/training_insights.json and accompanying CSV exports.

DESCRIPTIVE 02 / 05

T48 is consistently associated with RUL

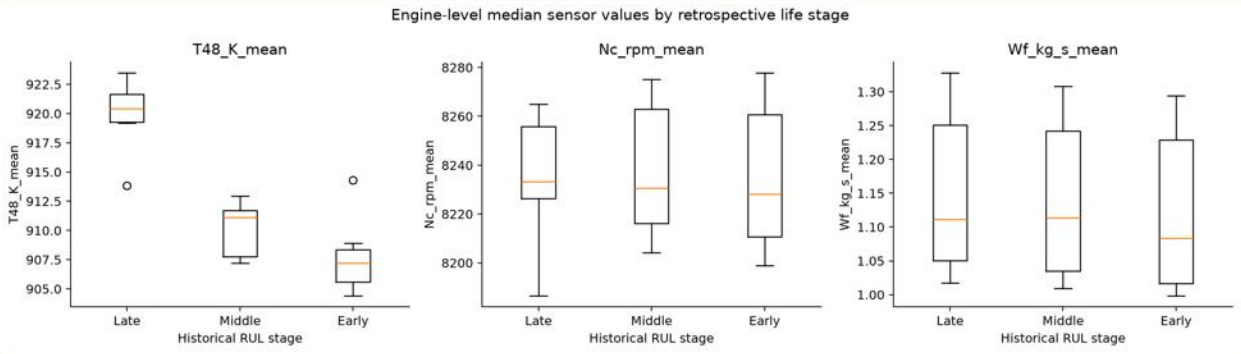


- Within-history Pearson r: -0.540 to -0.299.
- Negative in all 7 training source histories.
- Higher temperature tends to accompany lower historical RUL.
- Not a causal effect or prediction-accuracy result.

Evidence: Executed Task 5 notebook; reports/results/task5/training_insights.json and accompanying CSV exports.

DESCRIPTIVE 03 / 05

Historical stage groups tell a clear story

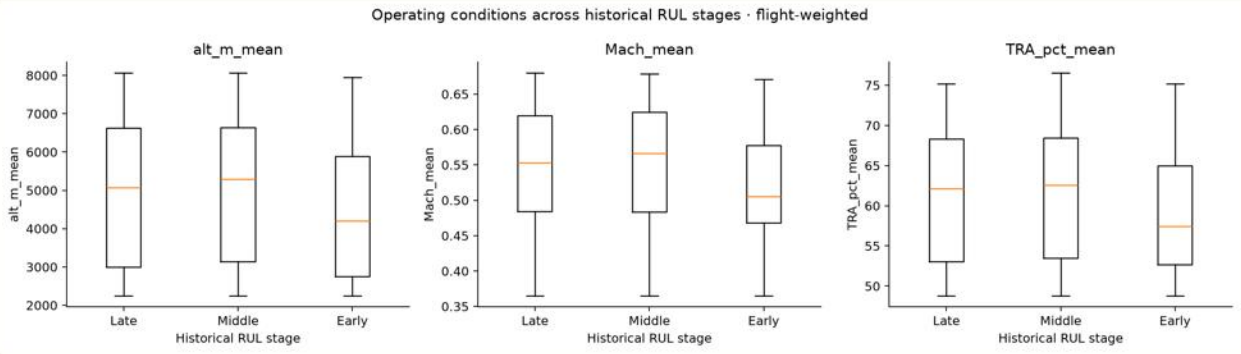


- Late 0–20 · Middle 21–50 · Early >50 cycles.
- Paired late−early engine-median T48: +13.62 K median.
- Range 4.93–17.82 K across 7 histories.
- Groups require true RUL; never use them as prediction inputs.

Evidence: Executed Task 5 notebook; reports/results/task5/training_insights.json and accompanying CSV exports.

DESCRIPTIVE 04 / 05

Context and heterogeneity temper the conclusion



- Nc–RUL correlations span −0.153 to +0.199.
- Operating-condition medians differ across RUL stages.
- Previous-flight changes show weak pooled linear associations.
- H2 added predictive value and H3 causal effects remain untested.

Evidence: Executed Task 5 notebook; reports/results/task5/training_insights.json and accompanying CSV exports.

DESCRIPTIVE 05 / 05

Observation → interpretation → next decision

- Observe: late-life T48 is higher in this historical sample.
- Interpret: temperature and mission context merit investigation.
- Decide next: compare baseline and simple regressors on held-out engines.
- Do not infer savings or maintenance actions from correlation alone.

Evidence: Executed Task 5 notebook; reports/results/task5/training_insights.json and accompanying CSV exports.

ALGORITHM / SELÇUK

Linear regression: a transparent starting model

- Predict RUL as an intercept plus weighted input features.
- Fit weights from training examples.
- Good first comparison: simple, inspectable coefficients.
- Proposed for next stage; not yet trained.

Evidence: https://scikit-learn.org/stable/modules/linear_model.html#ordinary-least-squares (accessed 11 Nov 2026).

ALGORITHM / SELÇUK

How we will check the linear model

- Start with 20 baseline columns and constant mean/median references.
- Use development-engine validation; fit preparation in each fold.
- Report MAE/RMSE, per-engine errors and optimistic RUL.
- Check extrapolation, correlated predictors and residual patterns.

Evidence: https://scikit-learn.org/stable/modules/linear_model.html#ordinary-least-squares (accessed 11 Nov 2026).

ALGORITHM / DENZEL

Random forest regression: average many trees

- Trees split feature space into rule-based regions.
- Average predictions across randomized trees.
- Candidate for nonlinear relations and interactions.
- Proposed comparison, not an established improvement.

Evidence: https://scikit-learn.org/stable/modules/ensemble.html#forest (accessed 11 Nov 2026).

ALGORITHM / DENZEL

Keep the forest comparison modest

- Predeclare a small depth/leaf-size search on development groups.
- Compare identical targets, splits and metrics with the baseline.
- Watch overfitting, collection effects and limited extrapolation.
- Feature importance would not prove causality.

Evidence: https://scikit-learn.org/stable/modules/ensemble.html#forest (accessed 11 Nov 2026).

NEXT STEPS

The next decision is a fair model comparison

- 12 Nov progress; rehearse for the 19 Nov milestone.
- After milestone: constants + linear regression + random forest.
- Then: grouped validation, limitations, engine-wise errors.
- Final target: 21 January 2027, per existing course plan.

Evidence: Supplied Project Tasks V6.0, pp. 1–2 (9 Nov 2026); milestone comments handout.

READINESS / LIMITS

What we can defend—and what remains open

- Verified: data structure, label logic, preparation and descriptive findings.
- Limits: small fleet, regime confounding, download-version match, cross-subset independence.
- Now confirmed: P2 is fan-inlet total pressure (journal Table 5).
- Unknown: predictive accuracy, real-world performance and cost savings.

Evidence: Arias Chao et al. (2021), Data 6(1), 5. https://doi.org/10.3390/data6010005

SOURCES / QUESTIONS

Questions to carry into the next stage

- Can the descriptive signal improve held-out-engine RUL estimates?
- How much is operating context versus degradation?
- Would a simpler DS01-only model be more defensible?
- Sources and AI assistance are documented in the repository.

Evidence: Arias Chao et al. (2021), Data 6(1), 5. https://doi.org/10.3390/data6010005